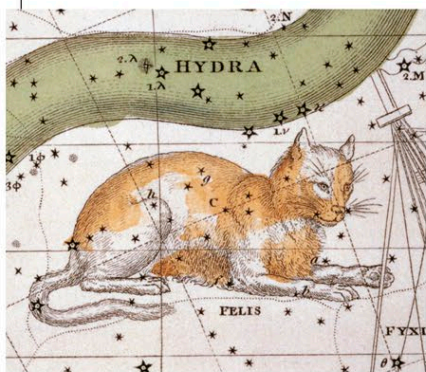


CONSTELLATIONS



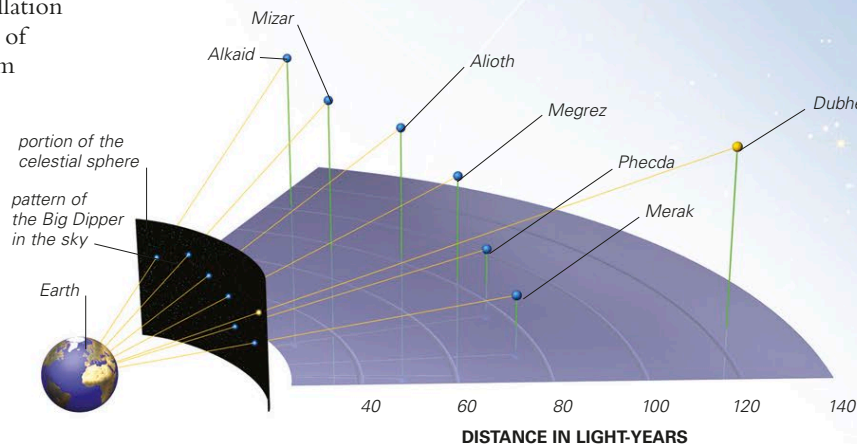
LOST CONSTELLATIONS

Some constellations have proved short-lived. In the 19th century, *Felis*, the cat, was incorporated into what is now part of the constellation of Hydra. It appeared on several star charts but was not officially adopted.

of a constellation area belong to that constellation, even if they are not connected to the stars that produce the constellation figure. Within some constellations are some smaller, distinctive groups of stars known as asterisms; these include Orion's belt (a line of three bright stars in Orion) and the Big Dipper (a group of seven stars in the constellation Ursa Major). A few asterisms cut across constellation boundaries. For example, most of the "Square of Pegasus" asterism is in Pegasus, but one of its corners is in Andromeda.

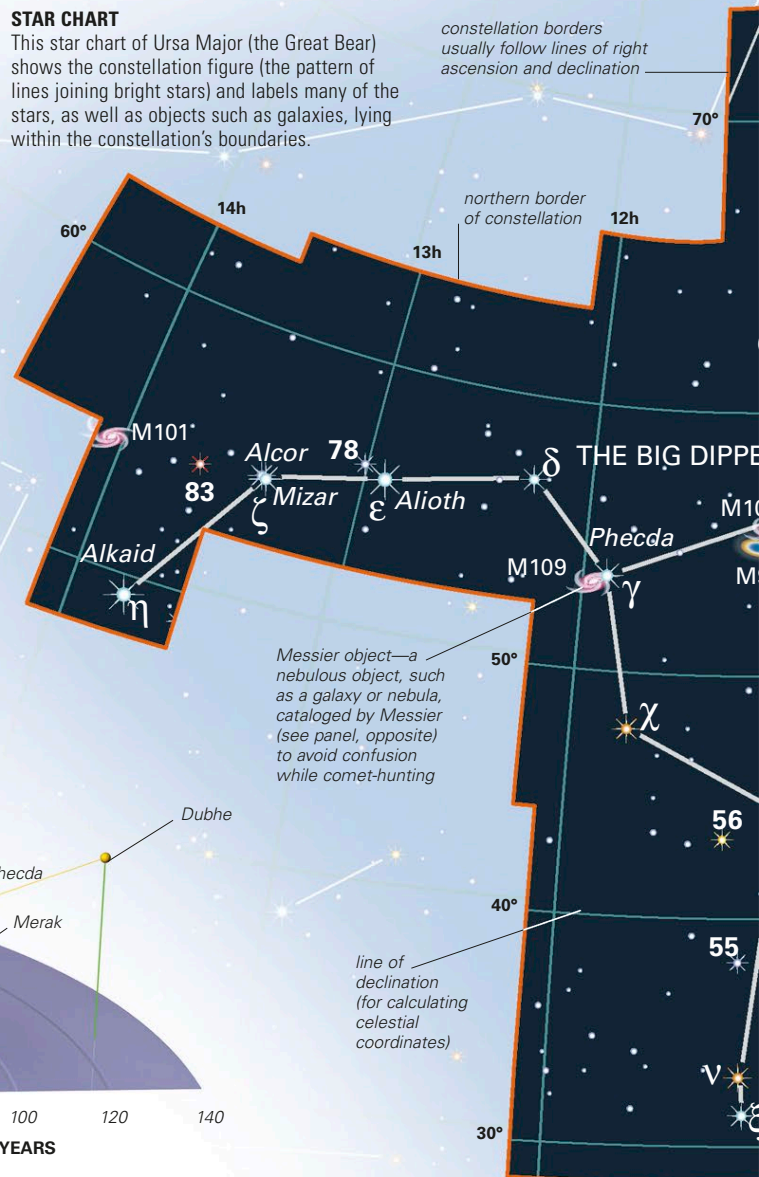
LINE-OF-SIGHT EFFECT

A star pattern such as the Big Dipper in Ursa Major is a two-dimensional view of what may be a widely-scattered sample of stars. The stars might seem to lie in the same plane, but they are at different distances from Earth. If we could view the stars from elsewhere in space, they would form a totally different pattern.



STAR CHART

This star chart of Ursa Major (the Great Bear) shows the constellation figure (the pattern of lines joining bright stars) and labels many of the stars, as well as objects such as galaxies, lying within the constellation's boundaries.



EXPLORING SPACE

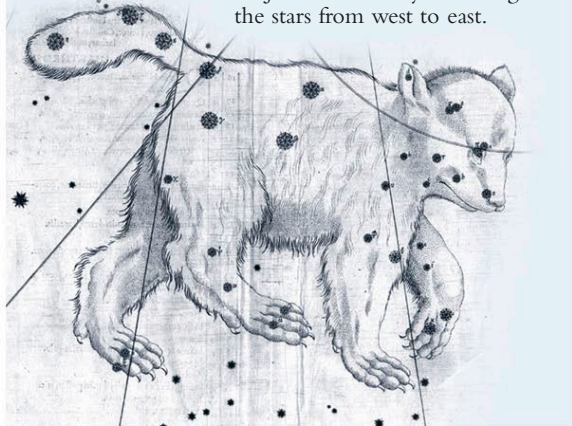
BAYER'S SYSTEM

Johann Bayer ascribed Greek letters to the stars in a constellation, roughly in order of decreasing brightness. Regulus, the brightest star in the constellation of Leo,

BAYER'S MAP OF URSA MAJOR

The seven stars of the Big Dipper can be seen in the upper left area of this chart from Bayer's *Uranometria*.

was given the name Alpha (α) Leonis, the second brightest (Denebola) was called Beta (β) Leonis, and so on. In some cases, Bayer used other ordering systems. The Big Dipper in Ursa Major is lettered by following the stars from west to east.

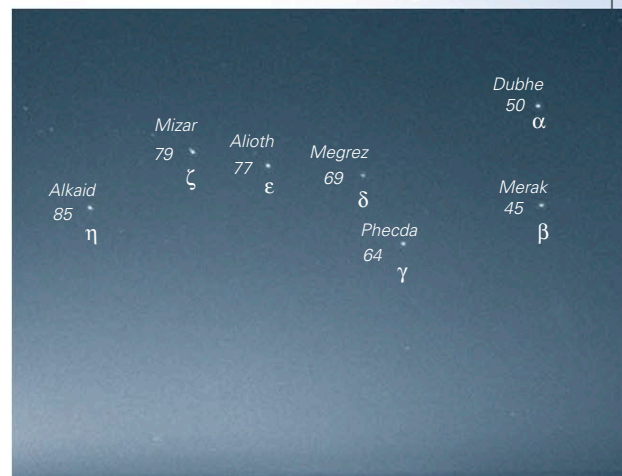


NAMING THE STARS

Most of the brighter stars in the sky have ancient names of Babylonian, Greek, or Arabic origin. The name Sirius, for example, comes from a Greek word meaning "scorching."

The first systematic naming of stars was introduced by Johann Bayer in 1603 (see panel, left, and p.347).

Bayer distinguished up to 24 stars in each constellation by labeling them with Greek letters, after which he resorted to using Roman lowercase letters, a to z. In 1712, English astronomer John Flamsteed (1646–1719) introduced another system, in which stars are numbered in order of their right ascension (see p.63) from west to east across their constellation. Stars are usually named by linking their Bayer letter or Flamsteed number with the genitive form (possessive case) of the constellation name—so 56 Cygni denotes the star that is 56th closest to the western edge of the constellation Cygnus. Since the 18th century, numerous further catalogs have identified and numbered many more faint stars, and specialized systems have been devised for cataloging variable, binary, and multiple stars.



SYSTEMS OF BAYER AND FLAMSTEED

This photo of the Big Dipper in Ursa Major shows the ancient name of each star, its Bayer designation, and its Flamsteed number. For example, the star Alkaid can also be called Eta (η), Ursa Majoris (Bayer), or 85 Ursa Majoris (Flamsteed).